The Delegated Eye

Scanning independence falls with alarm tenure while compliance holds perfect

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Background

Automated drowning-detection alarms promise a second set of eyes on the water. Whether lifeguards keep scanning independently once an alarm watches too, or hand the watching over to it, is untested outside vendor demonstrations.

Aims

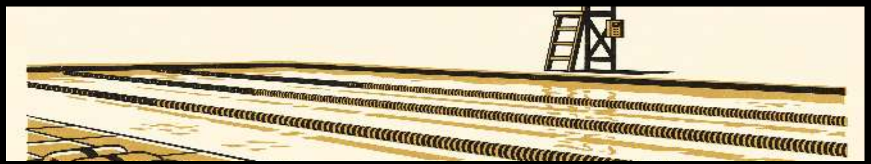
- Chart independent scan frequency against alarm tenure.
- Test whether response-time compliance stays perfect as scanning declines.
- Check whether independence recovers during a scheduled outage.

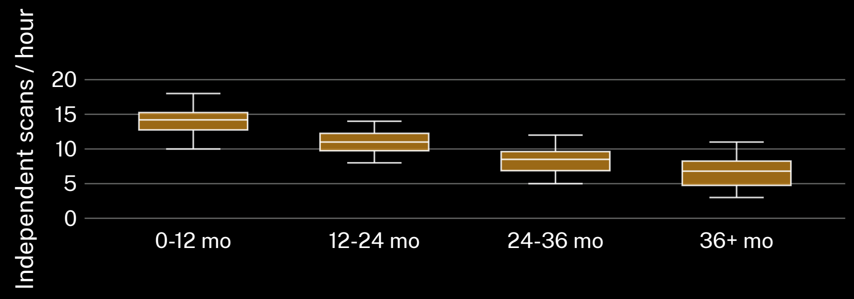
Materials & methods

- n = 58 lifeguards · 9 pools · overhead scan-logging · 6-month rolling window
- Scans coded from footage at 3 random intervals/shift, coder blind to alarm status
- Response-time compliance audited separately; 4-second threshold held constant

Findings

Independent scan frequency falls as alarm tenure lengthens. Measured response-time compliance holds at ceiling across the same window.





Median independent scans fell from 14.2 to 6.8/hour between lifeguards' first and third-plus year on an installed alarm (n = 58, 9 pools, March-August 2025); logged response time held under 4 seconds in 100% of audited events throughout.

Discussion & conclusions

Response time is audited; independent scanning is not — and the two move in opposite directions as tenure lengthens. Next: a pre-registered trial randomising scheduled alarm downtime by shift.

References

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Ethics approval 2025/162; footage reviewed on a rolling delete schedule, no patron identities retained.

“The alarm never blinks. Increasingly, neither does the log.”

Independent scans per lifeguard fell 52% across three years under the alarm (n = 58).